6.CCNP Enterprise作业六——IPv6 基础

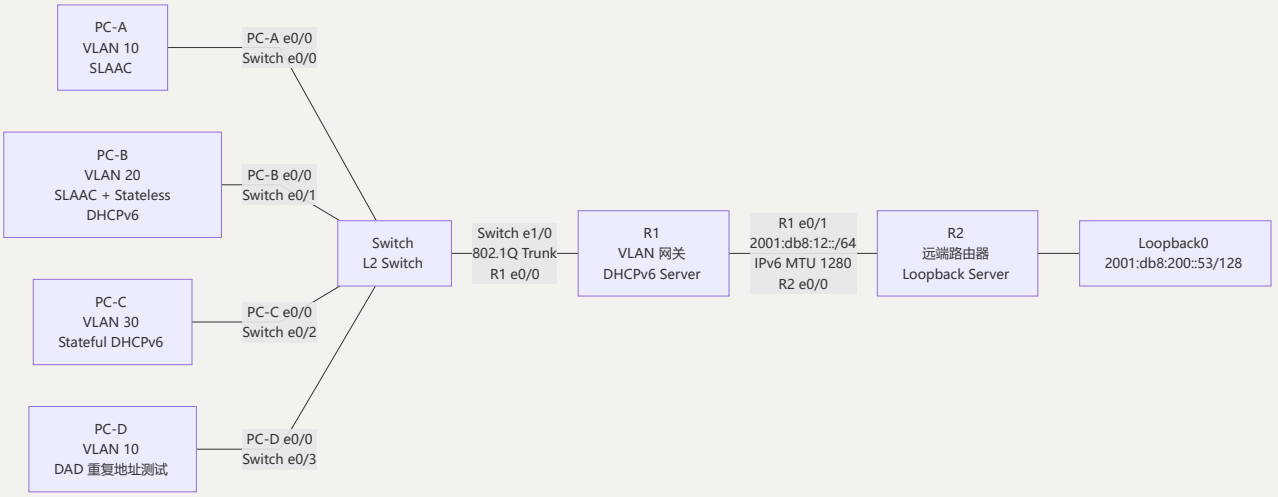
作业主题

IPv6 基础、NDP、SLAAC 与 DHCPv6

作业目标

1. 掌握 IPv6 地址、前缀和 Link-Local 地址的基础配置
2. 理解 IPv6 中 RS、RA、NS、NA 与 NDP 的工作过程
3. 掌握 SLAAC、Stateless DHCPv6、Stateful DHCPv6 的配置与验证方法
4. 掌握 IPv6 静态路由的配置与验证方法
5. 理解 IPv6 中 DAD、NUD 与 PMTUD 的基本现象

实验拓扑



设备角色规划

设备	作用
R1	VLAN 10、VLAN 20、VLAN 30 的默认网关，同时作为 DHCPv6 Server
R2	远端 IPv6 路由器，提供 Loopback0 测试地址
Switch	二层交换机，划分 VLAN 10、VLAN 20、VLAN 30
PC-A	VLAN 10 客户端，通过 SLAAC 获取 IPv6 地址
PC-B	VLAN 20 客户端，通过 SLAAC 获取地址，通过 Stateless DHCPv6 获取其他参数
PC-C	VLAN 30 客户端，通过 Stateful DHCPv6 获取 IPv6 地址
PC-D	VLAN 10 客户端，用于验证 DAD 重复地址检测

接口连接规划

连接	本端接口	对端接口	说明
PC-A 到 Switch	PC-A E0/0	Switch E0/0	VLAN 10，SLAAC only
PC-B 到 Switch	PC-B E0/0	Switch E0/1	VLAN 20，SLAAC + Stateless DHCPv6
PC-C 到 Switch	PC-C E0/0	Switch E0/2	VLAN 30，Stateful DHCPv6
PC-D 到 Switch	PC-D E0/0	Switch E0/3	VLAN 10，DAD 重复地址测试
Switch 到 R1	Switch E1/0	R1 E0/0	802.1Q Trunk，承载 VLAN 10、20、30
R1 到 R2	R1 E0/1	R2 E0/0	三层点到点链路
R2 到 Loopback	R2 Loopback0	-	远端 IPv6 测试地址

接口及 **IPv6** 地址规划

网段	用途	前缀	网关或地址
VLAN 10	SLAAC only	`2001:db8:100:10::/64`	`2001:db8:100:10::1`
VLAN 20	SLAAC + Stateless DHCPv6	`2001:db8:100:20::/64`	`2001:db8:100:20::1`
VLAN 30	Stateful DHCPv6	`2001:db8:100:30::/64`	`2001:db8:100:30::1`
R1-R2	点到点链路	`2001:db8:12::/64`	R1 `2001:db8:12::1`，R2 `2001:db8:12::2`

网段	用途	前缀	网关或地址
R2 Loopback0	远端测试地址	`2001:db8:200::53/128`	`2001:db8:200::53`

VLAN 与接入方式规划

VLAN	终端	**SWITCH** 接口	地址获取方式	**RA / DHCPV6** 要求
VLAN 10	PC-A	E0/0	SLAAC only	RA 通告前缀，`M=0`，`O=0`
VLAN 10	PC-D	E0/3	手工配置重复地址	用于触发 DAD 失败
VLAN 20	PC-B	E0/1	SLAAC + Stateless DHCPv6	RA 通告前缀，`M=0`，`O=1`
VLAN 30	PC-C	E0/2	Stateful DHCPv6	RA 设置 `M=1`，地址由 DHCPv6 分配

静态路由规划

设备	目的网段	下一跳地址
R1	`2001:db8:200::/64`	`2001:db8:12::2`
R2	`2001:db8:100::/48`	`2001:db8:12::1`

实验观察规划

实验项目	观察对象
SLAAC	PC-A 获取 IPv6 地址、默认网关与 RA 前缀
Stateless DHCPv6	PC-B 通过 SLAAC 获取地址，通过 DHCPv6 获取 DNS 等其他参数
Stateful DHCPv6	PC-C 通过 DHCPv6 获取 IPv6 地址，默认网关仍来自 RA
DAD	PC-D 与 PC-A 配置重复地址后，观察重复地址检测结果
NDP	观察 NS、NA、邻居表项生成与邻居状态变化
PMTUD	R1-R2 链路 IPv6 MTU 设置为 1280，观察 Packet Too Big
IPv6 静态路由	PC-A、PC-B、PC-C 到 R2 Loopback0 的跨网段通信

配置需求

1. 根据实验拓扑完成设备连接与基础配置
2. 在 Switch 上创建 VLAN 10、VLAN 20、VLAN 30
3. 将 Switch 的 E0/0、E0/3 配置为 Access 端口并加入 VLAN 10
4. 将 Switch 的 E0/1 配置为 Access 端口并加入 VLAN 20
5. 将 Switch 的 E0/2 配置为 Access 端口并加入 VLAN 30
6. 将 Switch 的 E1/0 配置为 Trunk，并允许 VLAN 10、VLAN 20、VLAN 30 通过
7. 在 R1 上启用 IPv6 单播路由功能
8. 在 R1 的 E0/0 上配置 Router-on-a-Stick 子接口，分别作为 VLAN 10、VLAN 20、VLAN 30 的 IPv6 网关
9. 在 R1 上配置 VLAN 20 的 Stateless DHCPv6 服务
10. 在 R1 上配置 VLAN 30 的 Stateful DHCPv6 服务
11. 在 R1、R2 的点到点链路上配置 IPv6 地址，并将 IPv6 MTU 设置为 1280
12. 在 R2 上配置 Loopback0 地址 `2001:db8:200::53/128`
13. 在 R1、R2 上配置 IPv6 静态路由，实现各 VLAN 与 R2 Loopback0 的互通
14. 按照规划分别完成 PC-A、PC-B、PC-C、PC-D 的 IPv6 客户端配置
15. 验证 SLAAC、Stateless DHCPv6、Stateful DHCPv6、DAD、NDP、PMTUD 与 IPv6 静态路由现象

作业要求

1. 请给出 R1、R2、Switch、PC-A、PC-B、PC-C、PC-D 的最终配置，并按照设备进行分类

R1

```
!
! Last configuration change at 11:25:54 UTC Mon Oct 19 2026
!
version 15.7
```

```
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
```

```
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
ipv6 dhcp pool v20-stateless
dns-server 2001:DB8:200::53
domain-name ccnp.lab
!
ipv6 dhcp pool v30-stateful
address prefix 2001:DB8:100:30::/64
dns-server 2001:DB8:200::53
domain-name ccnp.lab
!
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!
!
redundancy
!
no cdp log mismatch duplex
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
interface Ethernet0/0
no shutdown
no ip address
duplex auto
!
interface Ethernet0/0.10
no shutdown
description vlan10-slaac-only
encapsulation dot1Q 10
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:100:10::1/64
!
interface Ethernet0/0.20
no shutdown
description vlan20-slaac-stateless-dhcpv6
encapsulation dot1Q 20
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:100:20::1/64
ipv6 nd other-config-flag
ipv6 dhcp server v20-stateless
!
interface Ethernet0/0.30
no shutdown
description vlan30-stateful
encapsulation dot1Q 30
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:100:30::1/64
ipv6 nd prefix 2001:DB8:100:30::/64 no-advertise
ipv6 nd managed-config-flag
ipv6 dhcp server v30-stateful
!
interface Ethernet0/1
no shutdown
description r1-to-r2e0/0
no ip address
duplex auto
ipv6 address FE80::1 link-local
```

```
ipv6 address 2001:DB8:12::1/64
ipv6 mtu 1280
!
interface Ethernet0/2
no shutdown
no ip address
duplex auto
!
interface Ethernet0/3
no shutdown
no ip address
duplex auto
!
interface Ethernet1/0
no shutdown
no ip address
duplex auto
!
interface Ethernet1/1
no shutdown
no ip address
duplex auto
!
interface Ethernet1/2
no shutdown
no ip address
duplex auto
!
interface Ethernet1/3
no shutdown
no ip address
duplex auto
!
interface Ethernet2/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/1
```

```
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/3
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/3
no shutdown
no ip address
shutdown
duplex auto
```

```
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 route 2001:DB8:200::/64 2001:DB8:12::2
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
transport input none
!
!
end

```

R2

```
!
```

```
! Last configuration change at 11:25:52 UTC Mon Oct 19 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R2
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!
redundancy
!
no cdp log mismatch duplex
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Loopback0
no shutdown
```

```
description remote-test-server
no ip address
ipv6 address 2001:DB8:200::53/128
!
interface Ethernet0/0
no shutdown
description r2-to-r1-e0/1
no ip address
duplex auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:12::2/64
ipv6 mtu 1280
!
interface Ethernet0/1
no shutdown
no ip address
duplex auto
!
interface Ethernet0/2
no shutdown
no ip address
duplex auto
!
interface Ethernet0/3
no shutdown
no ip address
duplex auto
!
interface Ethernet1/0
no shutdown
no ip address
duplex auto
!
interface Ethernet1/1
no shutdown
no ip address
duplex auto
!
interface Ethernet1/2
no shutdown
```

```
no ip address
duplex auto
!
interface Ethernet1/3
no shutdown
no ip address
duplex auto
!
interface Ethernet2/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/3
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/1
no shutdown
```

```
no ip address
shutdown
duplex auto
!
interface Ethernet3/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/3
no shutdown
no ip address
shutdown
duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 route ::/0 2001:DB8:12::1
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
```

```
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
transport input none
!
!
end

```
SWITCHT

```
!
! Last configuration change at 11:25:51 UTC Mon Oct 19 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW1
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
```

```
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
no shutdown
switchport access vlan 10
switchport mode access
spanning-tree portfast edge
!
interface Ethernet0/1
no shutdown
switchport access vlan 20
switchport mode access
spanning-tree portfast edge
!
interface Ethernet0/2
no shutdown
switchport access vlan 30
switchport mode access
```

```
spanning-tree portfast edge
!
interface Ethernet0/3
no shutdown
switchport access vlan 10
switchport mode access
spanning-tree portfast edge
!
interface Ethernet1/0
no shutdown
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet1/1
no shutdown
!
interface Ethernet1/2
no shutdown
!
interface Ethernet1/3
no shutdown
!
interface Ethernet2/0
no shutdown
!
interface Ethernet2/1
no shutdown
!
interface Ethernet2/2
no shutdown
!
interface Ethernet2/3
no shutdown
!
interface Ethernet3/0
no shutdown
!
interface Ethernet3/1
no shutdown
```

```
!
interface Ethernet3/2
no shutdown
!
interface Ethernet3/3
no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
!
end

```

PC-A

```
!
! Last configuration change at 10:49:39 UTC Mon Oct 19 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname PC-10-SLAAC
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
no ip domain lookup
ip cef
no ipv6 cef
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!
!
redundancy
!
no cdp log mismatch duplex
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
no shutdown
no ip address
```

```
duplex auto
ipv6 address autoconfig default
ipv6 enable
!
interface Ethernet0/1
no shutdown
no ip address
duplex auto
!
interface Ethernet0/2
no shutdown
no ip address
duplex auto
!
interface Ethernet0/3
no shutdown
no ip address
duplex auto
!
interface Ethernet1/0
no shutdown
no ip address
duplex auto
!
interface Ethernet1/1
no shutdown
no ip address
duplex auto
!
interface Ethernet1/2
no shutdown
no ip address
duplex auto
!
interface Ethernet1/3
no shutdown
no ip address
duplex auto
!
interface Ethernet2/0
```

```
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/3
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/2
no shutdown
no ip address
shutdown
duplex auto
```

```
!
interface Ethernet3/3
no shutdown
no ip address
shutdown
duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
transport input none
!
!
end
```

```
### PC-B
```

```
!
! Last configuration change at 10:56:52 UTC Mon Oct 19 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname PC2-20-SLAAC+STARELESS
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
!
!
!
```

```
!
!
!
!
```

```
!
!
!
!
ip domain list ccnp.lab
no ip domain lookup
ip cef
no ipv6 cef
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!
!
redundancy
!
no cdp log mismatch duplex
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
interface Ethernet0/0
no shutdown
no ip address
duplex auto
ipv6 address autoconfig default
ipv6 enable
ipv6 dhcp client request vendor
!
interface Ethernet0/1
no shutdown
no ip address
duplex auto
!
interface Ethernet0/2
no shutdown
no ip address
duplex auto
!
interface Ethernet0/3
no shutdown
no ip address
duplex auto
!
interface Ethernet1/0
no shutdown
no ip address
duplex auto
!
interface Ethernet1/1
no shutdown
no ip address
duplex auto
!
interface Ethernet1/2
no shutdown
no ip address
```

```
duplex auto
!
interface Ethernet1/3
no shutdown
no ip address
duplex auto
!
interface Ethernet2/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/3
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/1
no shutdown
no ip address
```

```
shutdown
duplex auto
!
interface Ethernet3/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/3
no shutdown
no ip address
shutdown
duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
```

```
privilege level 15
logging synchronous
line vty 0 4
login
transport input none
!
!
end

```
### PC-C
```


!
! Last configuration change at 10:52:34 UTC Mon Oct 19 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname PC-30-STATEFUL
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
```

```
!
no ip icmp rate-limit unreachable
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
no ip domain lookup
ip cef
no ipv6 cef
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!
!
redundancy
!
no cdp log mismatch duplex
!
ip tcp synwait-time 5
!
!
```

```
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
no shutdown
no ip address
duplex auto
ipv6 address autoconfig default
ipv6 enable
ipv6 address dhcp
ipv6 dhcp client request vendor
!
interface Ethernet0/1
no shutdown
no ip address
duplex auto
!
interface Ethernet0/2
no shutdown
no ip address
duplex auto
!
interface Ethernet0/3
no shutdown
no ip address
duplex auto
!
interface Ethernet1/0
no shutdown
no ip address
duplex auto
!
```

```
interface Ethernet1/1
no shutdown
no ip address
duplex auto
!
interface Ethernet1/2
no shutdown
no ip address
duplex auto
!
interface Ethernet1/3
no shutdown
no ip address
duplex auto
!
interface Ethernet2/0
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet2/3
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/0
```

```
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/1
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/2
no shutdown
no ip address
shutdown
duplex auto
!
interface Ethernet3/3
no shutdown
no ip address
shutdown
duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
```

```
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
transport input none
!
!
end

```

PC-D
```

ASD
```

2. 请给出 PC-A、PC-B、PC-C 获取 IPv6 地址和默认网关的验证截图

PC-A

```
PC-10-SLAAC#show ipv6 int e0/0
Ethernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::A8BB:CCFF:FE00:100
  No Virtual link-local address(es):
  Stateless address autoconfig enabled
  Global unicast address(es):
    2001:DB8:100:10:A8BB:CCFF:FE00:100, subnet is 2001:DB8:100:10::/64 [EUI/CAL/PRE]
      valid lifetime 2591923 preferred lifetime 604723
  Joined group address(es):
    FF02::1
    FF02::1:FF00:100
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds
  Default router is FE80::1 on Ethernet0/0
PC-10-SLAAC#
```

PC-B

```
PC2-20-SLAAC+STARELESS#show ipv6 interface e0/0
Ethernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::A8BB:CCFF:FE00:200
  No Virtual link-local address(es):
  Stateless address autoconfig enabled
  Global unicast address(es):
    2001:DB8:100:20:A8BB:CCFF:FE00:200, subnet is 2001:DB8:100:20::/64 [EUI/CAL/PRE]
      valid lifetime 2591897 preferred lifetime 604697
  Joined group address(es):
    FF02::1
    FF02::1:FF00:200
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds
  Default router is FE80::1 on Ethernet0/0
PC2-20-SLAAC+STARELESS#
```

PC-C

```
Ethernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::A8BB:CCFF:FE00:300
  No Virtual link-local address(es):
  Stateless address autoconfig enabled
  Global unicast address(es):
    2001:DB8:100:30:E873:79DD:23AA:CB21, subnet is 2001:DB8:100:30:E873:79DD:23AA:CB2
1/128
  Joined group address(es):
    FF02::1
    FF02::1:FF00:300
    FF02::1:FFAA:CB21
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds
  Default router is FE80::1 on Ethernet0/0
PC-30-STATEFUL#
```

3. 请给出 R1 上 DHCPv6 Pool 和 DHCPv6 Binding 的验证截图

```
R1#SHOW IPV6 dhcp pool
DHCPv6 pool: v20-stateless
  DNS server: 2001:DB8:200::53
  Domain name: ccnp.lab
  Active clients: 0
DHCPv6 pool: v30-stateful
  Address allocation prefix: 2001:DB8:100:30::/64 valid 172800 preferred 86400 (1 in
use, 0 conflicts)
  DNS server: 2001:DB8:200::53
  Domain name: ccnp.lab
  Active clients: 1
R1#show ip
R1#show ipv
R1#show ipv6 dhc
R1#show ipv6 dhcp bi
R1#show ipv6 dhcp binding
Client: FE80::A8BB:CCFF:FE00:300
  DUID: 00030001AABBCC000300
  Username : unassigned
  VRF : default
  IA NA: IA ID 0x00030001, T1 43200, T2 69120
    Address: 2001:DB8:100:30:E873:79DD:23AA:CB21
            preferred lifetime 86400, valid lifetime 172800
            expires at Oct 21 2026 11:50 AM (171915 seconds)
R1#
```

4. 请给出 R1、R2 的 IPv6 路由表验证截图

R1

```
R1#show ipv6 route
IPv6 Routing Table - default - 10 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
C   2001:DB8:12::/64 [0/0]
     via Ethernet0/1, directly connected
L   2001:DB8:12::1/128 [0/0]
     via Ethernet0/1, receive
C   2001:DB8:100:10::/64 [0/0]
     via Ethernet0/0.10, directly connected
L   2001:DB8:100:10::1/128 [0/0]
     via Ethernet0/0.10, receive
C   2001:DB8:100:20::/64 [0/0]
     via Ethernet0/0.20, directly connected
L   2001:DB8:100:20::1/128 [0/0]
     via Ethernet0/0.20, receive
C   2001:DB8:100:30::/64 [0/0]
     via Ethernet0/0.30, directly connected
L   2001:DB8:100:30::1/128 [0/0]
     via Ethernet0/0.30, receive
S   2001:DB8:200::/64 [1/0]
     via 2001:DB8:12::2
L   FF00::/8 [0/0]
     via Null0, receive
R1#
```

R2

```
R2#SHOW IPV6 ROute
IPv6 Routing Table - default - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
S   ::/0 [1/0]
     via 2001:DB8:12::1
C   2001:DB8:12::/64 [0/0]
     via Ethernet0/0, directly connected
L   2001:DB8:12::2/128 [0/0]
     via Ethernet0/0, receive
LC  2001:DB8:200::53/128 [0/0]
     via Loopback0, receive
L   FF00::/8 [0/0]
     via Null0, receive
R2#
```

5. 请给出 PC-A、PC-B、PC-C ping `2001:db8:200::53` 成功的验证截图

PC-A

```
PC-10-SLAAC#ping ipv6 2001:db8:200::53
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:200::53, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
PC-10-SLAAC#
```

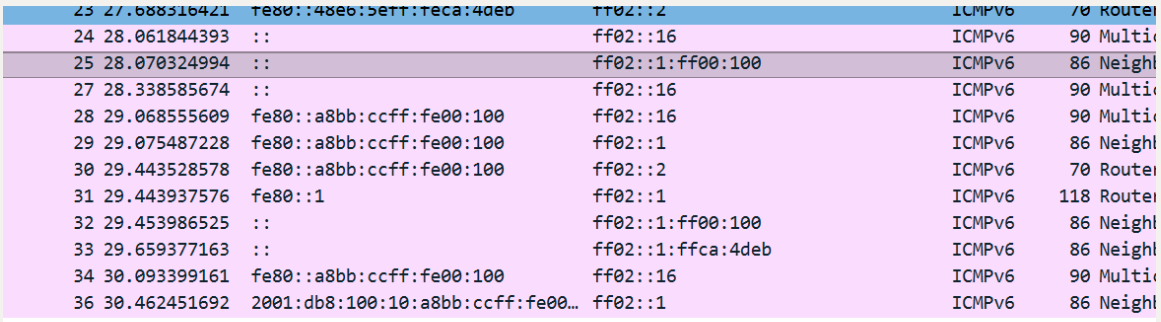
PC-B

```
PC2-20-SLAAC+STARELESS#ping ipv6 2001:db8:200::53
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:200::53, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
PC2-20-SLAAC+STARELESS#
```

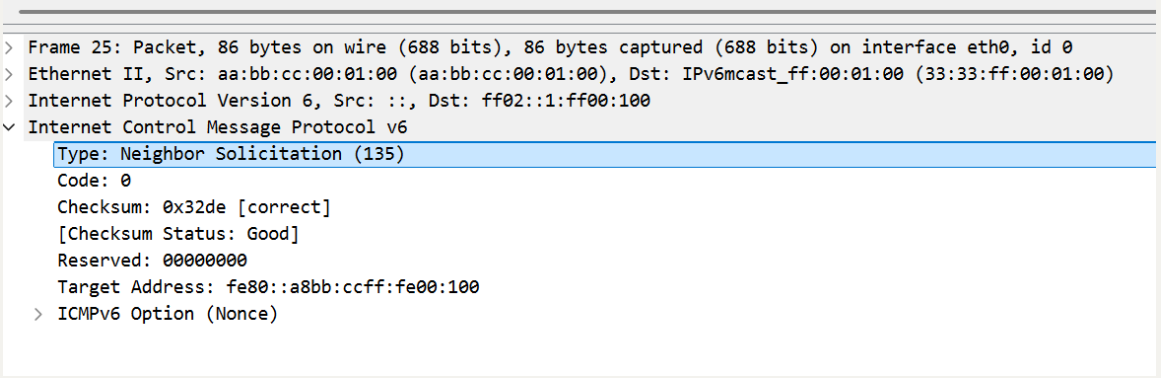
PC-C

```
PC-30-STATEFUL#ping ipv6 2001:db8:200::53
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:200::53, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/6/23 ms
PC-30-STATEFUL#
```

6. 请抓包或截图说明 PC-A 使用 SLAAC 时 RS、RA、DAD 报文的出现顺序

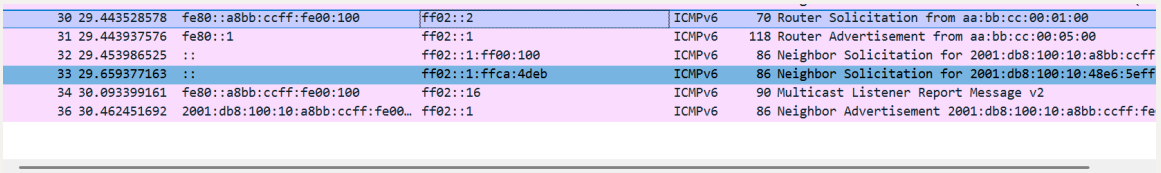


23	27.688316421	fe80::48e6:5eff:feca:4deb	ff02::2	ICMPv6	70 Router
24	28.061844393	::	ff02::16	ICMPv6	90 Multi
25	28.070324994	::	ff02::1:ff00:100	ICMPv6	86 Neigh
27	28.338585674	::	ff02::16	ICMPv6	90 Multi
28	29.068555609	fe80::a8bb:ccff:fe00:100	ff02::16	ICMPv6	90 Multi
29	29.075487228	fe80::a8bb:ccff:fe00:100	ff02::1	ICMPv6	86 Neigh
30	29.443528578	fe80::a8bb:ccff:fe00:100	ff02::2	ICMPv6	70 Router
31	29.443937576	fe80::1	ff02::1	ICMPv6	118 Router
32	29.453986525	::	ff02::1:ff00:100	ICMPv6	86 Neigh
33	29.659377163	::	ff02::1:ffca:4deb	ICMPv6	86 Neigh
34	30.093399161	fe80::a8bb:ccff:fe00:100	ff02::16	ICMPv6	90 Multi
36	30.462451692	2001:db8:100:10:a8bb:ccff:fe00…	ff02::1	ICMPv6	86 Neigh

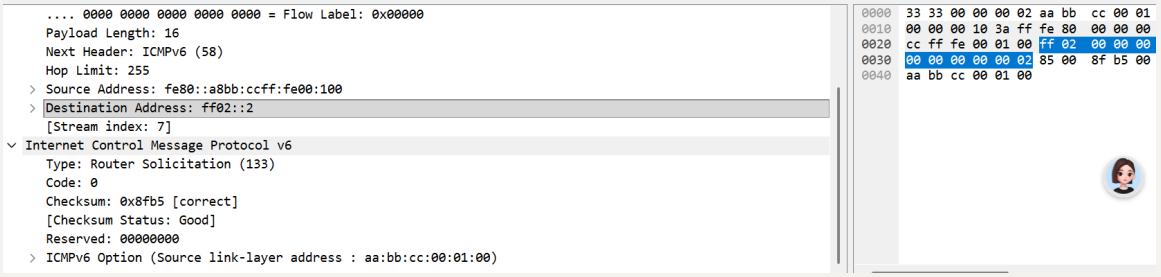


```
> Frame 25: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface eth0, id 0
> Ethernet II, Src: aa:bb:cc:00:01:00 (aa:bb:cc:00:01:00), Dst: IPv6mcast_ff:00:01:00 (33:33:ff:00:01:00)
> Internet Protocol Version 6, Src: ::, Dst: ff02::1:ff00:100
v Internet Control Message Protocol v6
    Type: Neighbor Solicitation (135)
    Code: 0
    Checksum: 0x32de [correct]
    [Checksum Status: Good]
    Reserved: 00000000
    Target Address: fe80::a8bb:ccff:fe00:100
  > ICMPv6 Option (Nonce)
```

主机 PC-A 生成Link-Local 地址并向组播地址ff02::1:ff00:100发送dad,确认链路本地地址可用



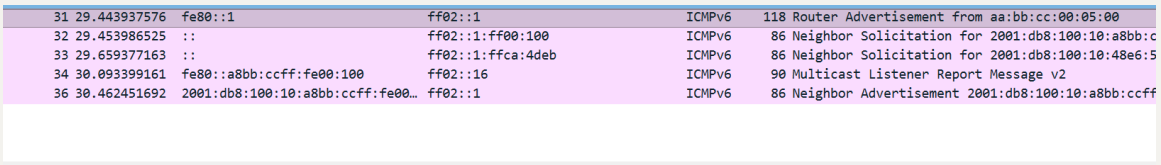
30	29.443528578	fe80::a8bb:ccff:fe00:100	ff02::2	ICMPv6	70 Router Solicitation from aa:bb:cc:00:01:00
31	29.443937576	fe80::1	ff02::1	ICMPv6	118 Router Advertisement from aa:bb:cc:00:05:00
32	29.453986525	::	ff02::1:ff00:100	ICMPv6	86 Neighbor Solicitation for 2001:db8:100:10:a8bb:ccff
33	29.659377163	::	ff02::1:ffca:4deb	ICMPv6	86 Neighbor Solicitation for 2001:db8:100:10:48e6:5eff
34	30.093399161	fe80::a8bb:ccff:fe00:100	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
36	30.462451692	2001:db8:100:10:a8bb:ccff:fe00…	ff02::1	ICMPv6	86 Neighbor Advertisement 2001:db8:100:10:a8bb:ccff:fe



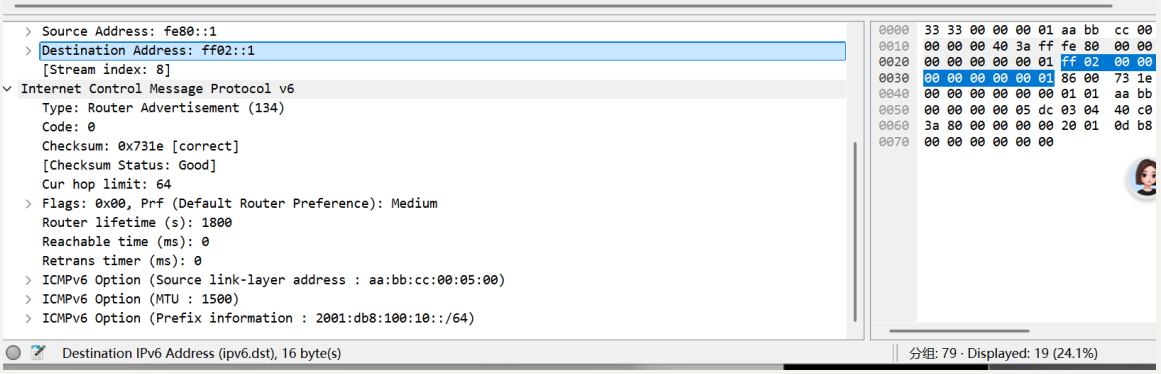
```
    .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
    Payload Length: 16
    Next Header: ICMPv6 (58)
    Hop Limit: 255
  > Source Address: fe80::a8bb:ccff:fe00:100
  > Destination Address: ff02::2
    [Stream index: 7]
v Internet Control Message Protocol v6
    Type: Router Solicitation (133)
    Code: 0
    Checksum: 0x8fb5 [correct]
    [Checksum Status: Good]
    Reserved: 00000000
  > ICMPv6 Option (Source link-layer address : aa:bb:cc:00:01:00)
```

```
0000  33 33 00 00 00 02 aa bb  cc 00 01
0010  00 00 00 10 3a ff fe 80  00 00 00
0020  cc ff fe 00 01 00 ff 02  00 00 00
0030  00 00 00 00 00 02 85 00  8f b5 00
0040  aa bb cc 00 01 00
```

主机用链路本地地址向路由器的广播地址fe02::2发送rs组播包



31	29.443937576	fe80::1	ff02::1	ICMPv6	118 Router Advertisement from aa:bb:cc:00:05:00
32	29.453986525	::	ff02::1:ff00:100	ICMPv6	86 Neighbor Solicitation for 2001:db8:100:10:a8bb:c
33	29.659377163	::	ff02::1:ffca:4deb	ICMPv6	86 Neighbor Solicitation for 2001:db8:100:10:48e6:5
34	30.093399161	fe80::a8bb:ccff:fe00:100	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
36	30.462451692	2001:db8:100:10:a8bb:ccff:fe00…	ff02::1	ICMPv6	86 Neighbor Advertisement 2001:db8:100:10:a8bb:ccff



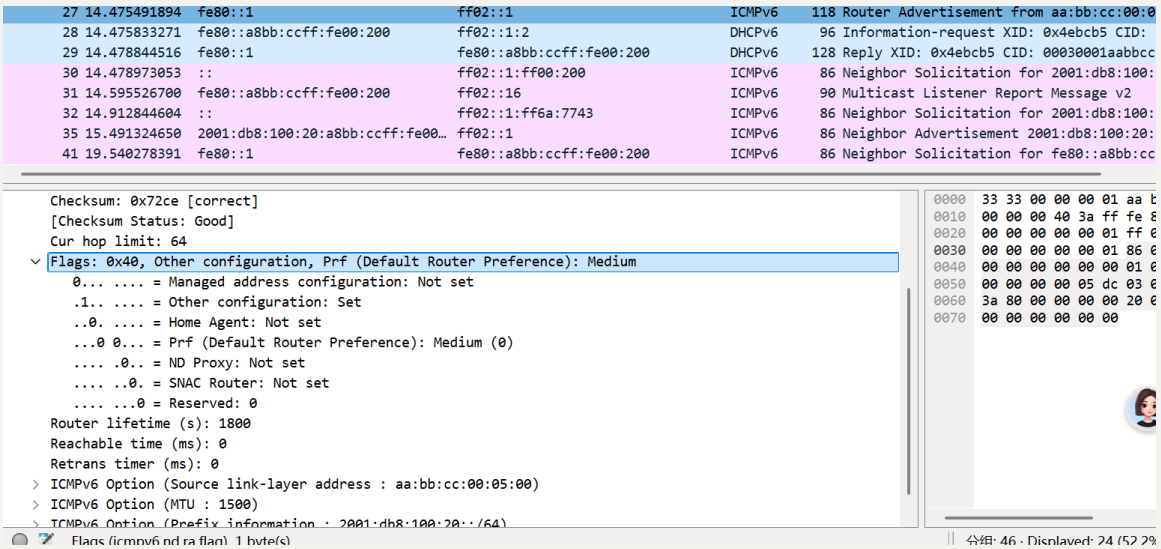
```
  > Source Address: fe80::1
  > Destination Address: ff02::1
    [Stream index: 8]
v Internet Control Message Protocol v6
    Type: Router Advertisement (134)
    Code: 0
    Checksum: 0x731e [correct]
    [Checksum Status: Good]
    Cur hop limit: 64
  > Flags: 0x00, Prf (Default Router Preference): Medium
    Router lifetime (s): 1800
    Reachable time (ms): 0
    Retrans timer (ms): 0
  > ICMPv6 Option (Source link-layer address : aa:bb:cc:00:05:00)
  > ICMPv6 Option (MTU : 1500)
  > ICMPv6 Option (Prefix information : 2001:db8:100:10::/64)
```

```
0000  33 33 00 00 00 01 aa bb  cc 00
0010  00 00 00 40 3a ff fe 80  00 00
0020  00 00 00 00 00 01 ff 02  00 00
0030  00 00 00 00 00 01 86 00  73 1e
0040  00 00 00 00 00 00 01 01  aa bb
0050  00 00 00 00 05 dc 03 04  40 c0
0060  3a 80 00 00 00 00 20 01  0d b8
0070  00 00 00 00 00 00
```

Destination IPv6 Address (ipv6.dst), 16 byte(s)　　分组: 79 · Displayed: 19 (24.1%)

路由器回包,并宣告前缀Prefix: 2001:db8💯10::

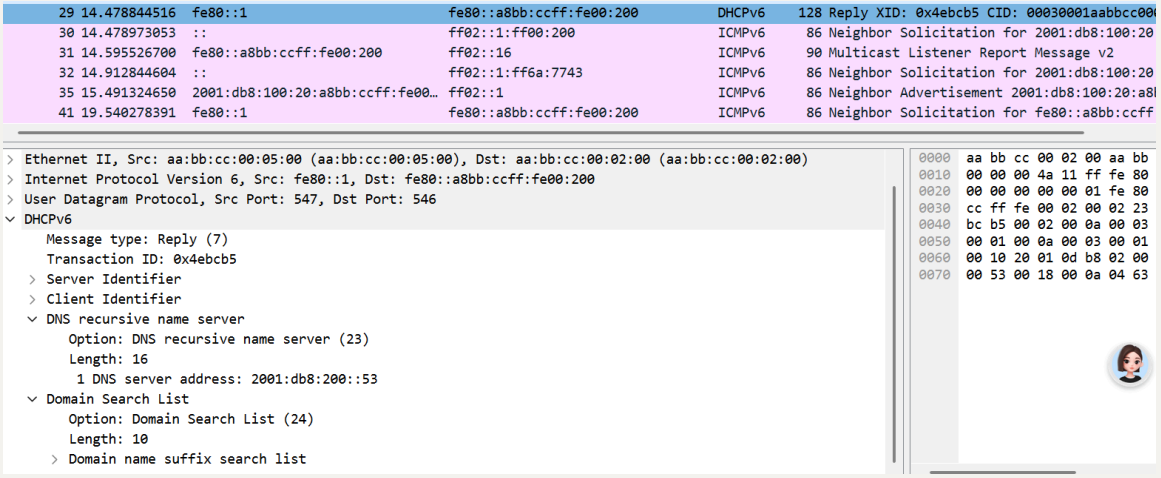
7. 请抓包或截图说明 PC-B 使用 Stateless DHCPv6 时 RA 与 DHCPv6 报文的作用



ra报文A Flag =1 告知主机该前缀允许SLAA

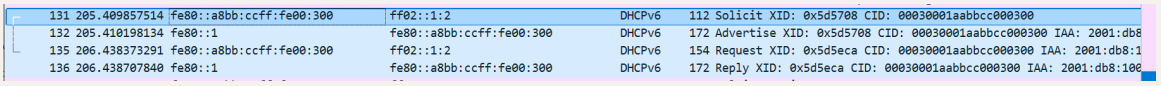
M Flag =0 不使用dhcpv6 获取地址

O Flag =1 告知使用dhcpv6 获取其他信息



dhcpv6告知dns地址和domain name

8. 请抓包或截图说明 PC-C 使用 Stateful DHCPv6 时地址分配过程



四个包对应ipv4的dora

首先通过ra报文的M Flag =1 使用dhcpv6 获取地址

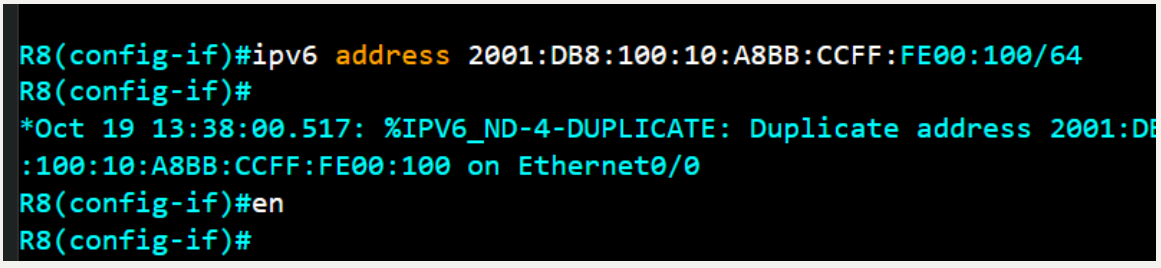
主机向fe02::1:2广播式问：“谁有能给我用的地址或参数？”

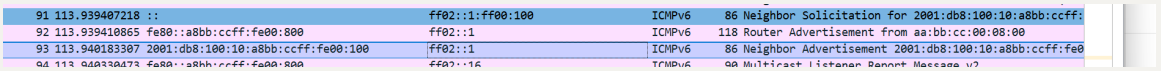
有资源的服务器应答：“我能给你，这是候选地址”

主机明确指定：“我选你（带 Server ID），请你正式分配”

正式下发租约，其他服务器看到 Server ID 就知道自己没被选中

9. 请通过 PC-D 制造 IPv6 重复地址，并说明 DAD 失败的现象和原因





pc4向申请地址的 Solicited-Node 组播，收到了pc1的na回复,所以失败

10. 请使用大包 ping 测试 PMTUD，并说明 ICMPv6 Packet Too Big 的触发原因

```
PC-30-STATEFUL#ping ipv6 2001:db8:200::53 size 1500
Type escape sequence to abort.
Sending 5, 1500-byte ICMP Echos to 2001:DB8:200::53, timeout is 2 seco
nds:
B!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
PC-30-STATEFUL#
```

R1-R2链路的IPv6 MTU被故意设置为==1280 ,从 PC-A发一个大于1280路径MTU的IPv6包到R2 Loopback,R1 会丢弃该包并返回Packet Too Big

11. 请说明 IPv6 中为什么不使用 ARP，以及 NDP 中 NS、NA 分别承担什么作用

arp它原有的一整套功能被重新设计并合并进了 ICMPv6/NDP,arp承载不了ipv6这么大的字段,arp使用广播,广播域的范围太大

NS:地址解析,DAD 查重,NUD 探测

NA:应答地址解析,应答 DAD,主动通告

12. 请说明 PC-A ping R2 Loopback0 的完整通信过程，包括默认网关获取、邻居发现、路由查表、Hop Limit 变化、二层封装变化与 Echo Reply 返回路径

阶段 0：PC-A 拿到地址和默认网关（SLAAC）：

a. R1 e0/0.10 起来后主动发 RA（Cisco 默认约 200s 周期，配置生效时会先发几次），目的地址 ff02::1（all-nodes），源地址=该子接口的链路本地地址 fe80::1。

b. PC-A 收到 RA 后：2001:db8:10::/64 + EUI-64 接口 ID → 2001:DB8💯10:A8BB:CCFF:FE00:100

c. DAD：PC-A 从 :: 向该地址的 solicited-node 组播 ff02::1:ff66:6800（MAC 层 33:33:ff:66:68:00）发 1 个 NS（ipv6 nd dad attempts 默认 1）。收不到别人回的 NA → 约 1 秒后变 preferred，开始可用。

d. 默认路由：PC-A 装 ::/0 → fe80::1，出接口 eth0。注意下一跳永远是链路本地地址，RA 不会给全局网关。

阶段 1：PC-A 查表 + 二层封装

a. 路由查找（最长匹配）：2001:db8:200::53 不匹配 2001:db8:10::/64（on-link），命中 ::/0 → 下一跳 fe80::1，出 eth0。

b. 构造 ICMPv6：Type 128（Echo Request）、Code 0、Identifier、

Sequence；载荷 56B（VPCS 默认）。IPv6 首部 40B，Hop Limit = 64，源=PC-A 全局地址。

c. 二层封装：dst MAC = R1 e0/0 的 MAC（不是 R2 的！），src MAC = PC-A，不打 VLAN tag（access 口发出的是无标签帧），EtherType 0x86DD。

阶段 2：R1 收包 → 转发（第 1 次 Hop Limit 递减）

a. e0/0 收到带 tag 的帧，因为配了 interface e0/0.10 + encapsulation dot1Q 10，R1 剥掉 VID=10 的 tag 交给子接口 .10 处理。

b. 目的 MAC = R1 e0/0 的 MAC → 交给 IPv6 处理；目的地址 2001:db8:200::53 不是本机地址 → 需要转发。

c. 查 FIB（show ipv6 route 输出应为）：2001:db8:200::53/128 via 2001:db8:12::2（静态）或走默认/汇总路由 → 出 e0/1。

d. Hop Limit 64 → 63（每台路由器转发减 1；若减到 0 就丢弃并回 ICMPv6 Type 3 Code 0 Time Exceeded）

阶段 3：R2 收到并产生 Echo Reply

a. R2 e0/0 收到，目的 MAC 是自己 → 交给 IPv6；目的地址 2001:db8:200::53 命中 Lo0 的 local 表项 → 上送处理。

b. 生成 ICMPv6 Type 129（Echo Reply），Code 0，Identifier/Sequence 原样回填，载荷同样 56B。

c. 新包：源 2001:db8:200::53，目的 2001:db8:10::250:79ff:fe66:6800，Hop Limit = 64（重新置默认值）。

阶段 5：Echo Reply 返回路径

a. R2 查表：目的 2001:db8:10::/64 没有直连 → 必须有回程路由（ipv6 route 2001:db8:10::/64 2001:db8:12::1 或 OSPFv3）。这是 90% “能通去程、不通回程/超时”的故障点：R2 上只写了 2001:db8:200::53/128 的静态路由而没有回程路由，ping 就会 100% 超时。

b. R2 → R1：NS 解析 2001:db8:12::1（或直接命中缓存），帧 src=R2 e0/0 MAC、dst=R1 e0/1 MAC，HL 64。

c. R1 收包：目的 2001:db8:10::... 是 .10 直连 → 命中 connected 路由，出 e0/0.10；HL 64 → 63。

d. R1 加 tag：e0/0.10 出方向按 encapsulation dot1Q 10 把 VID=10 打进

帧；目的 MAC 需要 PC-A 的——PC-A 地址在 .10 子网内，走 NS/NA（ff02::1:ff66:6800）后得到 PC-A MAC，缓存。

e. SW1：入 e1/0 带 tag 10 → 查 MAC 表命中 e0/0（access VLAN 10）→ 剥掉 tag，无标签帧发往 PC-A。

f. PC-A 收到：目的 MAC 自己、目的地址自己 → 匹配 Echo Reply 的 identifier/seq → 打印一行结果。VPCS 会显示 ttl=63（因为只经过 R1 一台路由器递减了一次）。